USN | | | | | | | | | | [22CSE287]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

INTERNET OF THINGS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	05 x02=10		
1.	a.	Define IoT and explain its significance in the context of modern digital ecosystems.	L-1	1	02
	b.	Discuss the relationship between IoT and digitization, emphasizing how IoT enhances digital transformation efforts.	L-1	2	02
	c.	Explain the role of security, networking and sensors in IoT.	L-1	3	02
	d.	Give 5 difference between Arduino and RaspberryPi?	L-4	4	02
	e.	What is the role of IoT for developing the smart Cities?	L-1	4	02
		PART-B			**06 X 05=30**
2.	a.	Explain the convergence of Information Technology (IT) and IoT, highlighting the synergies and challenges that arise from this integration.	L-2	1	05
		OR			
	b.	Identify and discuss three major challenges facing the widespread adoption and implementation of IoT solutions.	L-2	1	05
3.	a.	Discuss the role of IoT communication APIs in enabling interoperability and seamless data exchange between IoT devices and platforms, citing examples of commonly used APIs and their functionalities	L-2	2	05
		OR			
	b.	Explain the physical design considerations in IoT systems, including the role of IoT protocols in facilitating communication between devices and networks.	L-2	2	05
4.	a.	What are the main design principles that should be considered when designing an IoT architecture, and how do these principles contribute to scalability, flexibility, and security?	L-2	2	05
		OR			
	b.	Explain IoT architecture, including the layers, components, and their interconnections, emphasizing the flow of data and control within the system.	L-2	2	05

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5.	**a.**	Explain the concept of gateways in IoT, including their purpose in data management, protocol translation, and connectivity between edge devices and cloud services.	L-2	3	**05**
		OR			
	b.	Discuss the importance of data management in IoT applications, addressing challenges such as data storage, processing, analytics, and real-time insights generation.	L-2	3	**05**
6.	**a.**	Discuss the advantages and limitations of using Arduino UNO for IoT applications, considering factors such as power consumption, connectivity options, and available resources.	L-2	4	**05**
		OR			
	b.	Describe a real-world IoT project that could be implemented using both Arduino UNO and Raspberry Pi, and discuss the reasons for choosing each device in the project.	L-2	3	**05**
7.	**a.**	In the context of smart city use cases, explain the role of Industrial Automation and Commercial Building Automation.	L-2	4	**05**
		OR			
	b.	Compare and contrast the data representation and visualization techniques used in Industrial Automation and Commercial Building Automation within a smart city context.	L-2	4	**05**
		PART-C		**06 X 10=60**	
8.	**a.**	What are the main drivers behind the development of new network architectures for IoT, and how do they differ from traditional networking paradigms?	L-2	1	**10**
		OR			
	b.	Compare and contrast different IoT architectures such as centralized, decentralized, and hybrid models, discussing their strengths and weaknesses.	L-2	1	**10**
9.	**a.**	Describe the security considerations that are crucial in designing and implementing IoT architectures, addressing issues such as data privacy, authentication, access control, and secure firmware updates.	L-2	2	**10**
		OR			
	b.	Explain the importance of standards in IoT architectures, and discuss the role of standardization bodies such as IEEE, IETF, and ISO in shaping IoT protocols and interoperability.	L-2	2	**10**

10.	**a.**	Compare and contrast wide area networking (WAN) and local area networking (LAN) technologies used in IoT deployments, highlighting their strengths and limitations in different scenarios.	L-2	2	**10**
		OR			
	b.	How do local applications enhance the functionality of IoT systems, and what are the key considerations in designing and deploying such applications?	L-2	2	**10**
11.	**a.**	Outline the steps involved in setting up a Raspberry Pi-based IoT system, including hardware connections, software installation, and configuration of the operating system and networking components	L-2	2	**10**
		OR			
	b.	Discuss the role of libraries and external resources in enhancing the functionality of both Arduino and Raspberry Pi projects, with examples of popular libraries used in each platform.	L-2	2	**10**
12.	**a.**	Explain the key components of a smart city IoT architecture, including the roles of devices, networks, and interaction mechanisms.	L-2	3	**10**
		OR			
	b.	Analyze the security challenges faced by smart city systems and describe the essential elements of a smart city security architecture.	L-2	3	**10**
13.	**a.**	Provide an overview of a smart city use-case example, detailing the integration of various IoT components, and discuss the potential benefits and challenges of implementing such a system.	L-4	4	**10**
		OR			
	b.	Discuss the importance of interoperability in smart city IoT systems, and describe strategies and standards that can be employed to ensure seamless communication and data exchange among different devices and networks. Provide examples of relevant standards or protocols.	L-4	4	**10**